

# Supplementary Material for “Interaction Dynamics: A Physical-Interaction Reasoning Layer for Humanoid Robots”

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## I. CONTROLLER EQUATIONS AND PARAMETERS

The simulation runs at 500 Hz and the layer at 50 Hz ( $\Delta t = 0.02$  s). The planar-CoM error is  $e = c_{xy} - c_{xy}^{\text{ref}}$  with a deadband-shrunk error  $e_{\text{eff}} = e \max(0, \|e\| - \delta) / \|e\|$ . The external force  $\hat{F}_{\text{ext}}$  is estimated by the horizontal wrench estimator of the main paper, low-passed at 5 Hz.

**Confidence gate.** During quiescence ( $\|e\| < \delta$ ) an EWMA tracks the wrench-noise power,  $\sigma_f^2 \leftarrow (1 - \alpha)\sigma_f^2 + \alpha\|\hat{F}_{\text{ext}}\|^2$  with  $\alpha = 1 - e^{-2\pi f_\sigma \Delta t}$ , and  $F_{\text{cap}} = F_{\text{floor}} + k_\sigma \sqrt{\sigma_f^2}$ . The latch is

$$g \leftarrow \begin{cases} 1, & \|\hat{F}_{\text{ext}}\| > F_{\text{cap}}, \\ \max(0, g - \Delta t / \tau_g), & \|\hat{F}_{\text{ext}}\| \leq F_{\text{cap}} \wedge \|e\| < 1.5\delta, \\ g, & \text{otherwise,} \end{cases} \quad (1)$$

so capture stays engaged through a long recovery ( $\|e\| \geq 1.5\delta$ ) and releases only once recovered.

**Persistence.** A wrench timer, decaying at twice the fill rate,

$$t_F \leftarrow \begin{cases} t_F + \Delta t, & \|\hat{F}_{\text{ext}}\| > F_{\text{th}}, \\ \max(0, t_F - 2\Delta t), & \text{else,} \end{cases} \quad p = \text{clip}\left(\frac{t_F - \tau_0}{\tau_r}, 0, 1\right) \quad (2)$$

**Capture and hold.**  $a_e$  is the first input of a horizon- $N$  MPC on  $\ddot{e} = a_e + \hat{d}_{\text{eff}}$  minimizing  $\sum_j (q_p \|e_j\|^2 + q_v \|\dot{e}_j\|^2 + r \|a_{e,j}\|^2)$  s.t.  $\|a_e\| \leq a_{\text{max}}$ ; then  $u_{\text{cap}} = -k_{\text{map}} a_e$  (so  $k_{\text{map}}$ , in seconds, maps task acceleration to a velocity-command bias). The hold uses a  $p$ -gated leaky integral  $e_I \leftarrow 0.98 e_I + p e_{\text{eff}} \Delta t$ ,  $u_{\text{hold}} = -k_p e_{\text{eff}} - k_i e_I$ , and the output is

$$u = \text{sat}_{v_{\text{max}}} \left( \text{slew}[(1 - p)g u_{\text{cap}} + p u_{\text{hold}}] \odot s_{\text{axis}} \right), \quad (3)$$

added to the frozen policy’s walk command. Parameters are in Table I.

**Floor-corrected sustained drift.** For seed  $i$  and force  $F$ ,  $D_i(F) = |\bar{o}_i(F) - \bar{o}_i(0)|$ , where  $\bar{o}_i(F)$  is the settled lateral CoM offset from the recorded reference and  $\bar{o}_i(0)$  the same-seed no-force run; we report the median [IQR] over seeds. Same-seed subtraction cancels the gait-sway/reference-phase floor ( $\sim 30$  mm).

**Why a persistent bias does not fire the gate.** A constant foot-force bias  $b$  is absorbed by  $\sigma_f$ : during quiescence the EWMA drives  $\sigma_f^2 \rightarrow b^2$ , so  $F_{\text{cap}} \rightarrow F_{\text{floor}} + k_\sigma b \approx 16$  N at  $b = 5$  N, far above the bias—hence  $\|\hat{F}_{\text{ext}}\| = b$  never crosses  $F_{\text{cap}}$  and capture is not engaged. Sustained rejection is unaffected because the hold path is deviation-driven and ungated. A slowly-updated bias estimate  $\mu_F$  with the gate on

TABLE I  
LAYER PARAMETERS (AS IMPLEMENTED).

| symbol                        | meaning                  | value                         |
|-------------------------------|--------------------------|-------------------------------|
| $\Delta t$                    | control period           | 0.02 s (50 Hz)                |
| $\delta$                      | error deadband           | 0.03 m                        |
| $F_{\text{floor}}$            | force floor              | 1.0 N                         |
| $k_\sigma$                    | confidence margin        | 3.0                           |
| $f_\sigma$                    | noise-tracker bandwidth  | 0.3 Hz                        |
| $\tau_g$                      | gate-release window      | 1.6 s                         |
| $F_{\text{th}}$               | persistence threshold    | 3.0 N                         |
| $\tau_0, \tau_r$              | persistence delay, ramp  | 0.4 s, 0.4 s                  |
| $N$                           | MPC horizon              | 20 (0.4 s)                    |
| $q_p, q_v, r$                 | MPC weights              | 15, 100, 0.05                 |
| $a_{\text{max}}$              | MPC accel bound          | 6 m/s <sup>2</sup>            |
| $k_{\text{map}}$              | accel→cmd map            | 0.12 s                        |
| $k_p, k_i$                    | hold P, I                | 3.0, 8.0                      |
| $s_{\text{axis}}$             | ( $x, y$ ) channel scale | (0.2, 1.0)                    |
| $v_{\text{max}}, \text{slew}$ | bias clamp, rate limit   | 0.6 m/s, 6.0 m/s <sup>2</sup> |

**Algorithm 1** Confidence-gated capture–hold arbitration (per 50 Hz step)

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1:  $\hat{F}_{\text{ext}} \leftarrow m\ddot{c}_{xy} - \sum_i F_{i,xy}$   $\triangleright$  estimate; low-pass 5 Hz
2: if  $\|e\| < \delta$  then update  $\sigma_f^2$  (EWMA); return  $u = 0$   $\triangleright$  quiescent
3:  $F_{\text{cap}} \leftarrow F_{\text{floor}} + k_\sigma \sqrt{\sigma_f^2}$ ; update latch  $g$  by (1)
4: update persistence  $p$  by (2)
5:  $a_e \leftarrow \text{MPC}(e_{\text{eff}}, \dot{e}, \hat{d}_{\text{eff}})$ ;  $u_{\text{cap}} \leftarrow -k_{\text{map}} a_e$ 
6:  $u_{\text{hold}} \leftarrow -k_p e_{\text{eff}} - k_i e_I$   $\triangleright e_I$ :  $p$ -gated leaky integral
7:  $u \leftarrow (1 - p)g u_{\text{cap}} + p u_{\text{hold}}$ ; then  $\odot s_{\text{axis}}$ , slew-limit, saturate (3)
8: add  $u$  to the policy’s velocity command

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$\|\hat{F}_{\text{ext}} - \mu_F\|$  would separate bias from force explicitly; we note this as the cleaner design for hardware.